Apoorv Singh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.00
THE KHAITAN SCHOOL	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2021	96.80%
THE KHAITAN SCHOOL	Class X (CENTRAL BOARD OF SECONDARY EDUCATION)	2019	94.00%

SKILLS

- **Programming Languages:** Python, C++, C, SQL
- **Mathematics**: Probability & Statistics, Linear Algebra, Stochastic Differential Equations, Optimal Transport, Convex & Linear Optimisation
- **Modeling & Methods:** Langevin / Monte Carlo Sampling, Diffusion Models, Flow Matching, Schrödinger Bridges, Supervised Learning, Anomaly Detection
- **ML & Data**: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, SciPy, Matplotlib, Hugging Face Transformers
- **LLMs & Agents:** LangGraph, LangChain, RAG, Model Context Protocol (MCP), ChromaDB
- **Core CS & Tools:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Architecture, Linux, Git, Docker

EXPERIENCE

Machine Learning Intern | Galvanic Infotech

- Developed and benchmarked ML models (ANNs, SVM, Random Forest) for battery parameter estimation: State of Charge, State of Health, and remaining useful life prediction.
- Built fault and anomaly detection models flagging abnormal battery behavior from sensor data, optimizing inference pipelines in Python, TensorFlow, and Scikit-learn.

PROJECTS

M.Tech Thesis -- Constrained Generation via Schrödinger Bridges | Generative Modeling

Advisor: Prof. Saketha Nath Jagarlapudi, IIT Hyderabad -- Problem motivated by Fujitsu Research

- Formulated constrained sampling from frozen generative models as an entropy-regularized Schrödinger Bridge, proving convergence to the Wasserstein projection at rate $O(\epsilon^2)$.
- Built a score-based generative pipeline in d=100 and benchmarked proximal against Langevin correctors, showing the two trade off under matched budgets rather than one dominating.

Self-Correcting Text-to-SQL Agent | LangGraph, MCP, Groq, SQLite Link

- Built a LangGraph agent with a custom MCP server exposing five database tools over stdio, writing, running and self-repairing SQL by routing failures to targeted fixes.
- Raised the valid-query rate from 30% to 65% and tripled query accuracy on the BIRD Mini-Dev benchmark through error-class-conditioned repair.

AI Meeting Intelligence System | LangChain, Whisper, ChromaDB Link

- Engineered a RAG meeting assistant transcribing audio via Whisper, translating Hindi/Hinglish via Sarvam AI, and

answering queries using ChromaDB and sentence-transformers.
- Processed long recordings by splitting audio into 10-minute chunks with yt-dlp and FFmpeg, enabling transcription of hour-long meetings.
- Built a LangChain LCEL pipeline for map-reduce summarization, action-item extraction, and key-decision detection, with a chat interface.

Scientific Image Restoration | PyTorch, SSIM -- KLA AI Hack, IIT-H Link

- Architected FastSRNet, a UNet-style super-resolution network with residual blocks, reaching 0.773 SSIM on the competition leaderboard.
- Trained via a 3-phase curriculum (L1 warmup, combined L1/SSIM/perceptual/frequency loss, low-LR fine-tuning) with mixed-precision training and test-time augmentation.

SCHOLASTIC ACHIEVEMENTS

- Ranked in the **top 100 of 32,000+ teams** (top 0.3%) in the **Amazon ML Hackathon 2026**.
- Ranked in the **top 10** at the **KLA AI Hackathon 2026**, IIT Hyderabad.
- Ranked in the **top 4%** in CodeChef Starters 211.
- Secured **Rank 6** in the **IIT Hyderabad Deep Learning Hackathon** (image classification), with a model built from scratch without pretrained weights.
- Solved **1,200+ Data Structures & Algorithms problems** on LeetCode, CodeChef and GFG.

POSITIONS OF RESPONSIBILITY

- Placement Coordinator, Office of Career Services (OCS), IIT Hyderabad.
- Selected to represent the CS/AI department in front of leading Japanese companies at Hamamatsu Day, 2026.
- Teaching Assistant, IIT Hyderabad: Computer Architecture, Software Development, Advanced DSA.

RELEVANT COURSES

- **Theory Courses:** Advanced Data Structures & Algorithms, Linear Optimisation.
- **AI Courses:** Foundations of Machine Learning, Deep Learning, Visual Computing, Fraud Analytics Using Predictive and Social Network Techniques.
- **System Courses:** Advanced Computer Networks, Network Security.

EXTRACURRICULAR

- Black belt in karate; won a bronze medal at the state karate championship
- Performed as a guitarist in bands at music competitions.